«Spukhafte Fernwirkung»? Local Born geometry, the Tsirelson bound and the singlet curve: registered consistency checks, checkerboard interference and a retrocausal singlet model

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Abstract

Bell experiments without loopholes give CHSH values above the local bound 2, and high-precision measurements are compatible with Tsirelson's quantum bound 2√2 (Poh et al. 2015: 2.82759 ± 0.00051). This record asks what the author's framework, Mathematical Materiality (MM), must contain to respect both facts, and where it then stands among interpretations of quantum mechanics. No mathematical priority is claimed: the theorems used are published (Tsirelson; Barnum et al. 2010; Sorkin's quantum measures; Craig, Dowker et al. 2007; Aharonov's two-time formalism; Cramer; Argaman 2010; Wharton and Argaman 2020), and the record separates published results, derivations in the working notes and numerical conjectures. Predictions were registered with SHA-256 fingerprints before each test; failed predictions are reported. (1) Local readings and the Tsirelson bound. If each wing is read with the Born geometry (Euclidean state space with all its effects), no no-signalling joint event exceeds 2√2. Replacing the Euclidean ball with ℓ^p balls raises the maximum: 2^(2−1/p) for p ≥ 2 (the value of Ver Steeg and Wehner 2009), above 2√2 also for 1 < p < 2 (exact value conjectural). With marginals of length m the maximum is the quantum 2√(2 − m²). Two registered predictions failed: a hand formula, wrong for p < 2, and a three-dimensional search that stayed below the value already reached by the two-dimensional construction.

Version 2.48.2, 29 October 2026: one new section, 4.7, on the singlet correlation −cos θ from the two ends of the nexus (block 07_singoletto), a new title quoting Einstein's «spukhafte Fernwirkung», and rows added to sections 2, 7, 8 and 9. Everything else is the text of 2.48.1; the programs, results and registrations of the earlier blocks are unchanged.

Version 2.48.1, 26 October 2026: one new section, 4.6, on Born probabilities from counting copies of equal modulus (block 06_born_conteggio), with the sentences of sections 1, 2, 7, 8, 9 and 10 that it makes inaccurate or incomplete. Everything else is the text of 2.48; the programs, results and registrations of blocks 01–R3 are unchanged.

Version 2.48 of the Mathematical Materiality (MM) series, 13 October 2026. Continues MM 2.40 (the Feynman checkerboard, DOI 10.5281/zenodo.23280168). It also updates where MM stands among quantum interpretations (section 5), and corrects texts on quantum foundations published in versions 2.7–2.27 of *Mathematical Materiality: a status report* (concept DOI 10.5281/zenodo.21802475; section 6).

In the working records the material is in folders named 01_tsirelson, 02_copie_correlate, 03_loop_coerenza, 04_verbale_loop, 05_modello_loop, a second round R2 and R3; working files are in Italian.

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1. What this deposit is, and what it is not

It is **one link** in the MM programme, the quantum one. It shows how four working assumptions fit together (section 3), reports registered numerical consistency checks of each, and discusses their relation to published experimental results:

- local Born readings;
- weights built from pairwise correlations;
- a future boundary that selects the outcome (a "loop");
- the imaginary turn weight of the 2.40 checkerboard.

It is **not** a derivation of quantum mechanics, and it does not distinguish MM experimentally from standard quantum mechanics. For the Born rule, section 4.6 (new in 2.48.1) shows that in the 2.40 model the count of copies of equal modulus gives $|\psi|^2$ for rational ratios with H-es, H-eq and H-add; irrational ratios also require H-cont. The calculations cover denominators up to 12; the extension to arbitrary rational ratios uses the general construction in the cited sources. **No mathematical priority is claimed.** Three kinds of statement are kept apart below:

- published results (section 2);
- derivations given in the working notes, several supplied by the verifier;
- numerical conjectures.

Predictions labelled "registered" were written to a file and fingerprinted (SHA-256, UTC time) before the corresponding tests, according to the working records (section 9). Later findings are labelled as such. Predictions that failed are reported as failed (sections 4.1, 4.2, 4.3). Each block, and this text, was checked by an independent AI system (gpt-6-astra via Codex); its reports and the resulting changes are kept in the working records.

2. Prior work, first

topic	published result used here
local bound	Bell 1964; Clauser, Horne, Shimony, Holt, PRL 23, 880 (1969); Fine, PRL 48, 291 (1982)
quantum bound	Tsirelson (1980): $2\sqrt{2}$
no-signalling maximum	Popescu, Rohrlich, Found. Phys. 24, 379 (1994): S = 4
bipartite: full local quantum structure + no signalling ⇒ quantum correlations	Barnum, Beigi, Boixo, Elliott, Wehner, PRL 104, 140401 (2010)
local state-space geometry and non-locality	Janotta, Gogolin, Barrett, Brunner, NJP 13, 063024 (2011); Ver Steeg, Wehner, Quantum Inf. Comput. 9, 801 (2009), where the value 2^(2−1/p) already appears in their (different) p-norm theories; Bluhm, Jenčová, Nechita, CMP 393 (2022); Loulidi, Nechita, PRX Quantum 3, 040325 (2022)
tools	Clarkson, Trans. AMS 40, 396 (1936); Riesz, Acta Math. 49, 465 (1927) and Thorin, Kungl. Fysiogr. Sällsk. i Lund Förh. 8, 166 (1939) (Riesz–Thorin theorem); Horodecki R., P., M., Phys. Lett. A 200, 340 (1995)
quantum measure (grade 2)	Sorkin, Mod. Phys. Lett. A 9, 3119 (1994)
Born rule from envariance and counting (2.48.1, section 4.6)	Zurek, PRL 90, 120404 (2003); PRA 71, 052105 (2005); Phil. Trans. R. Soc. A 376, 20180107 (2018); objections: Schlosshauer, Fine, Found. Phys. 35, 197 (2005); Barnum, quant-ph/0312150 (2003); Mohrhoff, Int. J. Quantum Inf. 2, 221 (2004)
decision-theoretic and self-locating derivations (2.48.1)	Deutsch, quant-ph/9906015 (1999); Wallace, quant-ph/0312157 (2003) and arXiv:0906.2718 (2009); Saunders, Proc. R. Soc. A 460 (2004); Sebens, Carroll, Br. J. Philos. Sci. 69, 25 (2018); critical analysis of Deutsch's derivation: Barnum, Caves, Finkelstein, Fuchs, Schack, Proc. R. Soc. A 456, 1175 (2000)
Born rule from structure (2.48.1)	Busch, PRL 91, 120403 (2003); Masanes, Galley, Müller, Nat. Commun. 10, 1361 (2019)
Born square for radiative processes, transactional (2.48.1)	Kastner, Cramer, arXiv:1711.04501 (2018)
local model with a correlation linear in the angle (2.48.2, section 4.7)	Bell, Physics 1, 195 (1964)
$\cos^2$ law from statistical distinguishability or information (2.48.2)	Wootters, PRD 23, 357 (1981) and Entropy 15, 3130 (2013), arXiv:1301.2018; Summhammer, quant-ph/9910039; Brukner, Zeilinger, Found. Phys. 39, 677 (2009), arXiv:0905.0653
binary alternative, SU(2) and three-dimensional space (2.48.2)	von Weizsäcker, Aufbau der Physik (1985), reviewed by Lyre, quant-ph/0309183; Müller, Masanes, NJP 15, 053040 (2013)
spinors and null directions (2.48.2)	Penrose, Rindler, Spinors and Space-Time, vol. 1 (1984), cited from memory

joint quantum measures and Tsirelson	Craig, Dowker, Henson, Major, Rideout, Sorkin, J. Phys. A 40, 501 (2007): **strong** positivity gives Tsirelson; Barnett, Dowker, Rideout, J. Phys. A 40, 7255 (2007): without it PR boxes are allowed; Dowker, Henson, Wallden, NJP 16, 033033 (2014)
Hilbert space from a strongly positive decoherence functional	Dowker, Johnston, Sorkin, arXiv:1002.0589 (2010); it does not by itself give the dynamics, composite systems or frequencies
reconstructions	Chiribella, D'Ariano, Perinotti, PRA 84, 012311 (2011): purification together with other axioms
checkerboard and Dirac walk	Feynman, Hibbs (1965); Bisio, D'Ariano, Tosini (2015); MM 2.40
two-time formalism	Aharonov, Bergmann, Lebowitz, Phys. Rev. 134, B1410 (1964); Aharonov, Gruss, quant-ph/0507269 (2005); Aharonov, Cohen, Gruss, Landsberger, Quantum Stud. Math. Found. 1, 133 (2014); Gell-Mann, Hartle, gr-qc/9304023 (1994)
transactional interpretation	Cramer, Rev. Mod. Phys. 58, 647 (1986); Kastner, *The Transactional Interpretation of Quantum Mechanics* (Cambridge, 2012); Maudlin's contingent-absorber objection, *Quantum Non-Locality and Relativity*, 3rd ed. (Wiley-Blackwell, 2011)
retrocausal, locally mediated models	Costa de Beauregard, C. R. Acad. Sci. 236, 1632 (1953); Argaman, Am. J. Phys. 78, 1007 (2010); Price, arXiv:1002.0906 (2012); Leifer, Pusey, Proc. R. Soc. A 473, 20160607 (2017); Wharton, Argaman, Rev. Mod. Phys. 92, 021002 (2020); Almada et al., arXiv:1510.03706 (2016)
fine-tuning of no-signalling	Wood, Spekkens, New J. Phys. 17, 033002 (2015)
non-equilibrium and signalling	Valentini, Phys. Lett. A 156, 5 and 158, 1 (1991)
Local Friendliness	Bong et al., Nature Physics 16, 1199 (2020); Cavalcanti, Wiseman, arXiv:2106.04065 (2021)
self-consistency	Friedman, Morris, Novikov, Echeverria, Klinkhammer, Thorne, Yurtsever, Phys. Rev. D 42, 1915 (1990)

Published experimental and theoretical benchmarks.

- CHSH: Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$; Hensen et al., Nature 526, 682 (2015): 2.42 ± 0.20; Storz et al., Nature 617, 265 (2023): 2.0747 ± 0.0033.
- Third-order interference: Sinha et al., Science 329, 418 (2010): below 10^{-2} of the pairwise term; Kauten et al., NJP 19, 033017 (2017): more than four orders of magnitude below.
- Real versus complex quantum theory: Renou et al., Nature 600, 625 (2021) (theory); Li et al., PRL 128, 040402 (2022), more than 4.5 standard deviations in a network with two independent sources.
- Diósi–Penrose collapse: Donadi et al., Nature Physics 17, 74 (2021).

3. The working assumptions

They come from the author of the framework as premises; the exact wording is kept outside this deposit. Here they are in technical form, with what the translation adds.

- **W0 – local Born reading.** Each measured system is read with the Born rule of a two-level system. Its states form a Euclidean ball, the Bloch ball ($d = 3$) or its real disc ($d = 2$), with all its dual effects. As controls, section 4.1 replaces the Euclidean ball with real ℓ^p balls (state $\|r\|_p \le 1$, effects $\|u\|_q \le 1$, $1/p + 1/q = 1$). $B(p)$ denotes the maximal CHSH value for $d = 2$.
- **W1 – weights from copies correlated in pairs.** The weight of an event (a set of "copies") depends only on single copies and on pairs: $\mu(A) = \Sigma_{\alpha,\beta \in A} D_{\alpha\beta}$, a grade-2 measure in Sorkin's sense. *Added in translation:* D is positive semidefinite (a Gram matrix), so $\mu(A) = \|\Sigma_{\alpha \in A} v_\alpha\|^2$. The premise suggests, but does not impose, either step.
- **W2 – the loop.** The outcome written in a record is selected by the future end of its history; the history closes on itself. In section 4.3 this is a future boundary sampled with a measure. In section 4.4 it is a hidden parameter that depends on a future setting.
- **W3 – the imaginary turn weight.** On the 2.40 checkerboard a turn has weight $-i \sin \theta$ and a straight step $\cos \theta$ (θ = mass per tick).
- **Terminology.** In MM texts "non-local" means *not located in spacetime* (pregeometric). It refers to the underlying level of the framework: experimental events, settings and records are located in spacetime as usual. It does not mean influence outside light cones.

4. Tests and results

4.1 Local readings and the Tsirelson bound (01_tsirelson; registered 15:41:53 UTC)

Model. Two wings with the local spaces of W0. The joint event is any matrix C, with marginals, that gives non-negative probabilities for every pair of local effects (maximal tensor product, no signalling). For the unconstrained CHSH maximisation the marginals can be taken to be zero and the observables unbiased without loss of generality, and the constraint is then $\|C\|_{q \to p} \le 1$. The problem with marginals of fixed length m is treated separately below. H = [[1,1], [1,−1]].

Derivations (working notes, checked by the verifier).

- **Upper bound.** $S \le \|v_0+v_1\|_q + \|v_0-v_1\|_q \le 2^{(2-1/\max(p,q))}$, by Clarkson's inequalities, in any dimension.
- **p = 2.** The bound is 2√2 for every admissible joint event: the parallelogram law, i.e. Tsirelson's proof.
- **p ≥ 2.** The bound is attained by $C = 2^{(-1/p)}H$ (Riesz–Thorin with complex endpoints), so $B(p) = 2^{(2-1/p)}$.
- **1 < p < 2.** B(p) > 2√2 (proof supplied by the verifier). The upper bound is not attained. The best value found is the Hadamard construction $2^{(1+2/p)}/\|H\|_{q \to p}$; that it is the maximum is a conjecture.
- **Marginals of length m (p = 2, d = 2).** The maximum is exactly $2\sqrt{(2 - m^2)}$, the quantum value (proof supplied by the verifier). S = 2√2 forces both marginals to vanish.

CHSH maximum for d = 2. Proved for p ≥ 2 and at the endpoints; best values found numerically for 1 < p < 2.

p	1	1.1	1.25	1.5	1.75	1.9	2	2.1	2.5	3	4	8	∞
value	4	3.6526	3.3726	3.1065	2.9428	2.8699	2.8284	2.8755	3.0314	3.1748	3.3636	3.6680	4

Registered outcomes.

- **Passed:**
- a shared record read locally gives at most 2 (T1);
- 2√2 at p = 2 in d = 2 and 3 (T2);
- 4 at p = 1, ∞ (T3);
- marginals (T6);
- only p = 2 admits continuous norm-preserving rotations (T7: distortion ≥ $1.1 \cdot 10^{-2}$ for p = 1.5, 3, 4).
- **T4 failed for p < 2.** The numerical maxima lie 0.37–3.15 % below the hand formula $2\sqrt{2} \cdot 2^{|1/p-1/2|}$, which was registered before the computation. The hand calculation used the real value $\|H\|_{\infty \to 1} = 2$ in a complex Riesz–Thorin interpolation, where the complex value is 2√2.
- **T5 failed as written.** The random search in d = 3 fell below the d = 2 value. At p = 3 it fell short by $4.8 \cdot 10^{-4}$, outside the tolerance 10^{-4}: this case fails, and with it the test. At p = 1.5 it fell short by $9.4 \cdot 10^{-5}$ relative (inside the tolerance) or $2.9 \cdot 10^{-4}$ absolute (outside); at p = ∞ it matched.
- The registered tolerance did not say relative or absolute; the program used relative.
- The search did not evaluate the known construction unchanged.
- The second round R2-A shows that the embedded construction reproduces the d = 2 value to $3 \cdot 10^{-16}$. No d = 3 candidate exceeded it, which does not prove dimension independence for p < 2.

What it means. Within the stated bipartite maximal-tensor-product model, Euclidean local state spaces with their full dual effect spaces imply |S| ≤ 2√2: the cap comes from the local geometry, not from what correlates the wings. The measured S = 2.82759 ± 0.00051 is compatible. These data do not measure p: an ℓ^p reading would allow S above 2√2 without requiring it.

4.2 Two slits on the checkerboard (02_copie_correlate; registered 17:48:40 and 17:51:58 UTC)

Model. The 2.40 Dirac walk with θ = 0.5, slits at ±20 and a screen 1000 steps later. Three coins:

- the 2.40 coin, with $-i \sin \theta$ on turns;
- a real rotation coin;
- a coin with positive weights only.

Registered outcomes.

- **First run, passed:**
- positive amplitudes never give a negative cross term;

- the real coin reproduces the 2.40 probabilities to $3.3 \cdot 10^{-15}$ for one-direction sources and their mixture (G = diag(1, i) maps one coin to the other);
- third-order term $|I_3|/\max|I_2| = 8 \cdot 10^{-16}$ for the quadratic reading. In this statistic the variants |ψ|^p with p = 1 and 3 give 0.80 and 0.39; it is not the normalisation used by Sinha et al.;
- the cross-correlation matrix is positive (by construction).
- **First run, passed with limits:**
- T3b compares the same numerical vector in the two models, not corresponding preparations: it is uninformative;
- T5b compares the minimum fringe ratio over different regions.
- **First run, not met:** dark fringes (T2) and fringe spacing (T6). The registered "central region" contained 2 lattice sites, because the single-slit weight peaks at the ballistic front. The definition was at fault, not the code.
- **Second run, registered after seeing the first, with equal illumination of the two slits (all passed):**
- minimum ratio $\mu_{AB}/(\mu_A + \mu_B) = 4.1 \cdot 10^{-4}$;
- dark fringes at −143.2 and +140.4 sites (lattice sites −144 and 140), against ±142.1 from stationary phase with the lattice dispersion $\cos \omega = \cos \theta \cos k$;
- separation 283.6 against 284.3;
- with positive weights the ratio is never below 1.000000.

What it means.

- On the checkerboard every phase comes from the turns. At equal length, two more turns multiply the amplitude by $-\tan^2\theta$; between contributions to the same outcome this produces destructive interference. Positive amplitudes cannot produce a dark fringe.
- For the preparations tested here an equivalent real coin reproduces the probabilities. General comparisons need the corresponding transformation of preparations and measurements.
- Li et al. distinguish real and complex quantum theories in a network with independent sources, under the composition assumptions of that test; these computations do not address it.

Defect declared. The working diagnosis script imported the first program and rewrote its JSON file. The physical values are identical to the registered log; only the running time changed. The second round R2-C runs the same diagnosis in a temporary folder and gives the same output.

4.3 Can a future boundary give Born frequencies? (03_loop_coerenza; registered 18:08:51 UTC)

Model.

- The forward state on the screen comes from the checkerboard, with 8 screen intervals.
- The backward walker uses the adjoint step $U\dagger = C\dagger S\dagger$, whose coin carries $+i \sin \theta$ on turns.
- The future is a minimal record + environment space. Rules for selecting the future are compared by their total variation distance (TVD) from the Born weights w_b. The w_b are conditional on arrival in [−900, 900] and renormalised over the 8 intervals, like the frequencies.

Registered outcomes (all numerical thresholds met). Forward × backward equals the Born weight to $3 \cdot 10^{-16}$; this is an identity, not a physical selection.

rule for the future	TVD from Born
uniform (Haar) future weighted by loop closure $w_b \cdot \Phi_b$	^{2} 0.0010
definite-record futures, prior ∝ number of states d_b, weighted by closure	0.0069
uniform future, unweighted, outcome by the two-state (ABL) rule	0.099
uniform future, unweighted, largest closure wins	0.164
definite records, each record equally likely	0.489
definite records weighted by closure, each record equally likely (d_b unequal)	0.316
K independent copies of the record, largest closure wins (K = 1, 2, 4, 8, 16)	0.164, 0.042, 0.178, 0.285, 0.353

- The last row is not monotonic in K. The registered criterion (K = 16 farther than K = 1, K = 8 above 0.1) is met; the word "grows" in the prediction is not.
- **Protocol deviation.** The first run enlarged the two outer screen intervals. The values above come from the run with the registered intervals; 0.74 % of the weight falls outside the screen.

Found after the run, not registered: the exponential race.

- For independent equal-rate exponential variables E_b, choosing $\operatorname{argmin}_b(E_b/w_b)$ samples branch b with probability $w_b/\Sigma_c w_c$.
- The squared components $|\Phi_b|^2$ of a Haar-random complex vector (dimension ≥ number of branches) are such variables divided by a common sum. So "smallest $|\Phi_b|^2/w_b$ wins" reproduces Born (TVD 0.0024).
- It is an exact sampling algorithm and contains the weights.

What it means. In this minimal model a future boundary gives Born frequencies with some rules (closure-weighted, or the exponential race) and not with others. The measure over futures remains a postulate. In the prior-$\propto$-d_b rule the factor d_b cancels against the mean overlap $1/d_b$.

4.4 A locally mediated retrocausal singlet model (05_modello_loop; registered 19:17:49 UTC)

Model (an adaptation of Argaman 2010 to spin ½).

- The source is at (0, 0) of the checkerboard, the readings at $(\mp L, T)$. The choices are made at $T - 10$ and are space-like separated from the other wing's reading. The checkerboard supplies only the geometry of the paths; the angular law is assumed.
- Alice's future outcome fixes a parameter $\lambda \in \{a, a + \pi\}$ with probability p. The parameter is carried back along her path to the source, then forward along Bob's path as $\lambda + \pi$.
- Bob reads it with the local Born rule $\cos^2((\lambda_B - b)/2)$.
- In closed form, with $m = 2p - 1$ and $c = \cos(a - b)$: $P(A = u, B = v) = (1 + um)(1 - uvc)/4$. So $E = -c$ for every p, $\langle A \rangle = m$ and $\langle B \rangle = -mc$.
- At $p = ½$ these are the singlet joint probabilities for coplanar projective measurements.
- For other p, E is unchanged but the marginals allow signalling.

Registered outcomes (all passed).

- E within 0.0042 of $-\cos(a - b)$ on a 12×12 grid (threshold 0.008), and $S = -2.8281$.
- No observable signalling at $p = ½$: 0.0039 and 0.0040 (threshold 0.008).
- Signalling equal to $|2p - 1|$ for p = 0.6, 0.75, 0.9: 0.202, 0.502, 0.801 (tolerance 0.01).
- A forward-only control with sign responses gives $|S| = 2.0000$.
- Geometric path check passed (with $L = T$ the paths are straight).
- The symmetric mixture (either wing's future chooses) matches $-\cos(a - b)$ within 0.0037.

Second round R2-B (registered 19:30:20 UTC, all passed).

- Born responses with λ uniform and independent of the settings: $E = -½ \cos(a - b)$ within 0.0055, $|S| = 1.4157$.
- Real zigzags with $T > L$: λ is constant along every path in 20 000/20 000 histories.
- Screening: at fixed (λ_B, b), Bob's frequency does not depend on (a, A) (maximum 1.58 standard errors over 8 comparisons, threshold 5).
- Symmetric against asymmetric version, compared directly: within 0.0046 (threshold 0.012).

What it means.

- With future-input dependence $\rho(\lambda \mid a, b) \neq \rho(\lambda)$ and a local Born reading, the singlet correlations are reproduced. Every link of the chain runs along paths inside light cones, through the common past of the two readings.
- Within this family, no observable signalling holds if and only if the future chooses with the balanced (Born) weight. That link is already discussed by Argaman (2010), and the angular law $\cos^2$ is assumed.
- The cancellation at $p = ½$ is a fine-tuning in the sense of Wood and Spekkens.
- The parameter must not be readable at the source: read there, it would reveal a modulo π.
- Coverage: a singlet pair, coplanar projective measurements, ideal detectors.

4.5 Self-consistency and the balanced weight (R3; registered 20:09:57 UTC)

Question. In section 4.4 the balanced weight $p = ½$ is put in by hand. Why that weight? The author's premise answers

that this is not a "why" question: the existence of the present is what selects its past. Read technically, only self-consistent histories exist (Novikov; Friedman et al. 1990).

Computation (binomial distribution in floating point, no sampling).

- In the model of 4.4 with $p > ½$, Alice encodes a bit x in her setting (a = 0 or π, b = 0).
- N is odd, and the N pairs are assumed independent and identically distributed given x. Bob's frequency of B = +1 is then $½ \mp \varepsilon/2$, with $\varepsilon = |2p - 1|$, and he decodes x by majority. For $p < ½$ the decoder must be inverted; the grid uses $p \geq ½$.
- **Assumption, not computed:** a faster-than-light channel, composed with a second one in another frame, can bring the decoded bit back into Alice's past without further error. This is Tolman's "antitelephone" (Tolman, *The Theory of the Relativity of Motion*, 1917; Benford, Book, Newcomb, Phys. Rev. D 2, 263, 1970). The pair model does not provide this construction.
- **Paradox protocol:** Alice sets x = NOT(decoded bit). A history is consistent only if Bob decoded NOT x.
- W(N, p) = P(decoded bit = NOT x) is the weight of consistent histories under the distribution *before* imposing consistency.

p	N = 1	11	101	1001	10001
0.50 (Born)	0.5	0.5	0.5	0.5	0.5
0.51	0.490	0.473	0.420	0.263	0.0227
0.55	0.450	0.367	0.156	$7.6 \cdot 10^{-4}$	$5.9 \cdot 10^{-24}$
0.60	0.400	0.247	0.0209	$8.1 \cdot 10^{-11}$	$4.3 \cdot 10^{-91}$
0.75	0.250	0.034	$3.3 \cdot 10^{-8}$	$6.4 \cdot 10^{-65}$	underflow (positive, below double precision)

Registered outcomes (all passed).

- S1: with $p = ½$, $W = ½$ for every N in the grid.
- S2: for the values $p \neq ½$ tested, W decreases with N, below the registered bounds.
- S3: for every $p \neq ½$ tested, some N in the grid gives $W < 0.05$.

A general statement, proved afterwards (Hoeffding's inequality, supplied by the verifier): $W(N, p) \leq \exp(-N\varepsilon^2/2)$, so $W \to 0$ for every fixed $p \neq ½$.

What it shows, and what it does not.

- **Shown:** with any non-Born weight, a protocol with enough pairs makes the prior weight of consistent histories as small as one likes. With the Born weight the channel carries no information, so the antitelephone cannot be built from it at all, and W = ½ only as a formal comparison.
- **Not shown:** that a non-Born weight is *impossible*. For $0 < p < 1$ and every finite N, $W > 0$, so consistent histories always have non-zero weight (at p = 0 or 1, W = 0 already for one pair).
- **What the passage from "arbitrarily improbable" to "cannot exist" needs:** a further postulate. One formulation, close to the author's premise: *a present exists only if no protocol can make its consistency arbitrarily improbable*. Here it is checked only for the family of protocols described (inf over N of $W > 0$), with the return link and the resources assumed available for arbitrarily large N. It is stated as the author's postulate, not derived.
- **Another way to impose consistency:** condition on it. When $W > 0$, $P_C(h) = P(h) \cdot 1_C(h)/W$ gives the consistent histories total probability one, at the price of changing the statistics. The post-selected closed timelike curves of Lloyd et al. (PRL 106, 040403, 2011) exclude histories with zero post-selection amplitude and renormalise the others. Which measure describes the closed loop is part of the postulate.
- **Coverage:** one protocol, within the pair model; the local $\cos^2$ law is assumed throughout.
- **Prior art:** signalling outside the Born weights is known for pilot waves (Valentini, Phys. Lett. A 158, 1, 1991) and for this toy model (Argaman 2010). The reading "existence of the present ⇒ balanced weight" is the author's.
- **Registration:** the prediction file was written before the code; both were fingerprinted in the same minute, before the run.

4.6 Born probabilities from counting copies of equal modulus (06_born_conteggio; new in 2.48.1; registered 25–26 October 2026 UTC)

Question. With which declared assumptions does counting copies, in the 2.40 model, give the probability $p = |\psi|^2$, and where does the square enter?

The author's premises. Three sentences of the author, in Italian with their exact words, are in `documenti/registrations_original/06_born_conteggio/PREMESSE_UGO_RACCOLTE_20261026.md`:

- U1 (23 October 2026): if you have to describe a 10 by means of a number 1 (10 = 1), you are forced to distribute the descriptor 1 over 10 space-times of the number 10;
- U2 (13 October 2026): the rule is decided by the number and the mutual correlation of the copies of the nexus;
- U3 (13 October 2026): it is the future of a history that chooses its past, so the answer is the loop.

Their translations into mathematical objects are choices of the AI ("bridges" P1–P6 of the protocol), not the author's words.

Prior work, first. Twenty-two sources were read with the same eight questions (19 downloaded from arXiv, 3 already in the working records; they are not redistributed, their SHA-256 fingerprints are in `documenti/registrations_original/06_born_conteggio/IMPRONTE_FONTI.txt`). The derivations that count copies (Zurek 2003, 2005, 2018; the ancilla step of Deutsch 1999; Saunders 2004; Sebens and Carroll 2018) share three ingredients:

1. *equal modulus, equal probability*. This is the step the critics attack: for Schlosshauer and Fine (2005) it is an additional assumption about the connection between state and probabilities; for Barnum, Caves, Finkelstein, Fuchs and Schack (2000) Deutsch's version hides a principle of insufficient reason. Envariance gives a symmetry; equal probabilities need a further, contested link;
2. *fine-graining into copies of equal modulus, and counting*: an amplitude $\sqrt{(m/M)}$ becomes m copies of modulus $1/\sqrt{M}$ and the probability is m/M. None of the critics contests this step;
3. *the square comes from the conserved norm*, not from the count: Zurek attributes it to Pythagoras (the Euclidean geometry of Hilbert space); Saunders shows that with an evolution conserving an l^p norm the same argument would give |c|^p.

Decision-theoretic derivations (Deutsch; Wallace 2007, 2009) use other axioms (branching indifference, state supervenience, continuity). Structural routes do not count branches: Gleason and Busch (2003), Masanes, Galley and Müller (2019), Sorkin (1994). In the ABL rule of the two-state vector formalism the squared-amplitude probability is part of the rule. Kastner and Cramer (2018) present a derivation of the Born square for radiative processes in direct-action theory, from the requirement that a real photon involves both emission and absorption (their pp. 1 and 8-9); that the absorber's response carries the dual vector, and that the product is a probability, rest on the transactional picture (p. 2 and note 1). Both arguments are distinct from the copy counting of B-2. No source derives that probabilities exist, or which outcome occurs. Data: three-path interference is zero within $|\kappa| < 0.01$ (Sinha et al. 2010) and $\tilde{\kappa_3} = 0.0 \pm 3.1 \cdot 10^{-5}$ in the classical regime of Kauten et al. (2017).

Protocol and verification. Five formulations (B-1 counting paths without phases, a control; B-2 counting copies of equal modulus, U1; B-3 number and mutual correlation, U2; B-4 the loop, U3; B-5 which norm the coin conserves), with named assumptions:

- H-es: probabilities exist and depend only on the state of the copies;
- H-eq: copies of equal modulus are equiprobable;
- H-add: the probability of a set of distinct copies is the sum, normalised to 1;
- H-cont: continuity for irrational ratios;
- H-prod (B-4 only): the product of the forward and backward amplitudes is a probability.

The protocol was checked twice by a second AI system (gpt-6-luna via Codex), with 9 and 4 remarks, all accepted in two registered corrections before any computation; there was no third round. The verifier's main points became part of the classification: B-3 defines its measure as $|\Sigma w|^2$, which is already the Born form; B-4 uses a quadratic rule as a probability; the complex conjugate of the backward amplitude follows from defining the return with the adjoint step.

Tests and results (`codice/06_born_conteggio/conti_K1_K5.py`, registered before the run; the expected outcomes were written knowing the physics of quantum walks, so they are not blind predictions).

test	what it measures	result

K1, 2.40 coin	two slits, r = I_AB/(I_A + I_B), geometry of 4.2	min r = 4.1·10⁻⁴ (dark fringes; criterion min r < 0.05)		
K1, paths without phases	the same statistic with path counts	r = 1 at every site (to 2·10⁻¹⁶): no dark fringes		
K2	one coin step at θ = π/6, fine-graining into 4 cells	4 cells of modulus ½; counting gives 3/4 and 1/4 = \	ψ\	²
K2b	every ratio m/M with M ≤ 12 (66 cases)	M cells of modulus 1/√M; counting = \	ψ\	² to 9·10⁻¹⁶
K3	variation of Σ\	ψ\	^p over 200 steps, θ = π/6, π/4, π/3, 0.3	p = 2: ≤ 2.7·10⁻¹⁴; p = 1: 10.6–12.7; p = 1.5: 1.9–2.5; p = 3: 0.78–0.90; p = 4: 0.92–0.99
K4	Sorkin κ with three slits at −20, 0, +20	κ = 2.6·10⁻¹⁷ at the central maximum; inside the windows of both experiments		
K5	backward amplitude from the record	equal to the complex conjugate		
K5bis (added after the run, declared)	the same at five sites with imaginary amplitude	distance from the conjugate ≤ 8·10⁻¹⁷; from the amplitude itself 0.14–0.19		

In the registered K5 the record site (the maximum of the amplitude) had a real amplitude, where "conjugate" and "equal" coincide; K5bis repeats the check where they differ.

Classification.

formulation	outcome	where the square enters		
B-1 paths without phases	does not follow from the data (no dark fringes)	nowhere: the rule is linear		
B-2 copies of equal modulus	derived with H-es, H-eq, H-add in the cases tested (ratios with denominator up to 12); for any rational ratio the general construction of the sources applies (not redone here), and H-cont for irrational ones	in the number of cells, m_k ∝ \	ψ_k\	², because the coin conserves the sum of squares
B-3 number and correlation	assumes the quadratic measure (as Sorkin; K4 checks the program)	in the definition μ(A) = \	Σw\	²
B-4 the loop	circular for the Born rule; it remains a premise on the choice of the outcome	assumed: forward × backward is \	ψ\	² by the adjoint
B-5 the conserved norm	among the exponents, angles and state tested, only p = 2 stays constant	this is the point: the coin conserves the sum of squares and not the others		

What it means, as a reading.

- In the 2.40 model the count of copies of equal modulus gives $|\psi|^2$ for rational ratios with H-es, H-eq and H-add; irrational ratios also require H-cont. The calculations cover denominators up to 12; the extension to arbitrary rational ratios uses the general construction in the cited sources. The construction (fine-graining and counting) is analogous to Zurek's; here H-eq is declared as a premise, not derived from envariance, and the number of cells is read from the amplitudes.
- The square comes from the 2.40 coin together with the structure the model adopts: complex amplitudes, linear superposition, a coin that preserves the inner product. The same exponent appears in two separate results of section 4.1: in the stated model the $2\sqrt{2}$ cap is proved for p = 2; T7 concerns norm-preserving rotations and reports distortion for p = 1.5, 3 and 4.
- H-eq, the step contested in known physics, is in the framework the author's premise U1 (bridge P3, a choice of the AI): the objection does not disappear, it becomes a postulate.

- The 2.40 coin is the Feynman checkerboard: known physics adopted as a postulate.

Not claimed. That probabilities exist (H-es), which outcome occurs, a solution of the measurement problem. The tests are finite: one coin step for K2 and K2b, four angles and five exponents for K3. The general construction for arbitrary rational ratios is that of the sources.

4.7 The singlet correlation −cos θ from the two ends of the nexus (07_singoletto; new in 2.48.2; registered 29 October 2026 UTC)

The title quotes Einstein's «spukhafte Fernwirkung» (spooky action at a distance; letter to Max Born, 3 March 1947; the word «Verschränkung», entanglement, is Schrödinger's, 1935). The question mark is the point of this section: in the translations that hold, the mirror is a reading rule of one nexus with two ends, not a signal travelling between them, and the marginals stay zero, so nothing can be sent with it.

Question. Sections 4.1–4.6 bound the correlations (2 for a common record read locally, 2√2 for Born readings at the two ends) but never derive the curve that Bell experiments measure for two spin-½ particles in the singlet state, E(a, b) = −a·b = −cos θ. With the Born rule, a two-dimensional complex state space and rotations acting as SU(2), the curve follows in one line; the question is which of these ingredients can come from the two ends of the nexus, and which stay imported: (i) the exponent 2, (ii) an end as a two-state system with a phase, (iii) the three directions of space as directions of the end, with the half angle, (iv) the singlet as mirror reciprocity of the two ends, (v) zero marginals (no signalling).

Method. Six translations T0–T5, registered with criteria C1–C5 before the program was run (13:24:31 UTC): C1 E = −cos θ exactly or within 10^{-3} on a grid; C2 marginals within 10^{-3} of zero; C3 S = 2√2 at the standard angles and not above it on a grid of 64 directions; C4 dependence on the relative angle only, in the plane and for 200 random pairs of directions in space; C5 the list of what is imported. The expected outcomes had been worked out by hand before the registration: this is a declared contaminated preregistration, not a blind prediction. One program (`calcoli_livello1_c1.py`): the registered first version stopped on a rounding error (a ratio of −1e-17 raised to the power ½); the corrected file, which clips that ratio to [0, 1], was registered before it was run (correction 1).

Results (deviations written as 0 are below 10^{-15}):

translation	C1	C2	C3 (standard / grid maximum)	C4 plane / space	outcome	imported		
T0 quantum mechanics (control)	0	0	2.828427 / 2.828427	0 / 0	holds, everything imported	(i)–(v)		
T1 the 2.40 coin read by two oriented readers; 24 variants (k ∈ {½, 1}, phase χ ∈ {0, π/2, π}, weight \	amplitude\	^p with p = 1–4)	0 only for k = ½, χ = π, p = 2	0	2.828427 for that variant	0 / 0 for that variant	holds with imported ingredients (1 variant of 24)	p = 2; a two-component space with complex amplitudes and a tensor-product state; the sign χ = π; k = ½ and the azimuth; normalisation of the weights
T2 mirror: after the reading of A, the end B lies along −u·a and is read with a law f	—	0	—	—	does not determine the curve	—		
T2 with f = (1 + c)/2	0	0	2.828427 / 2.828427	0 / 0	holds with the law of the end imported	f		
T2 with f linear in the angle	0.2071	0	2 / 2	0 / 0	fails (Bell's straight line)	—		
T2 with f = cos²(3θ/2)	1.5364	0	2.828427 / 2.828427	0 / 0	fails (−cos 3θ)	—		

T3 copies counted with their noise (Wootters' route) + mirror	0	0	2.828427 / 2.828427	0 / 0	holds with imported ingredients	independent copies (binomial variance), linearity of mixtures, Fisher metric, equal distinguishability for equal rotations, end points, monotonicity, isotropy in 3D, fair first outcome, mirror
T4a three sides (2+1), probability = pair product normalised on the opposite direction	0	0	2.828427	0 / not defined	fails for non-coplanar settings (holds in the plane)	the pair rule, mirror
T4b four sides (3+1), same rule	0	0	2.828427 / 2.828427	0 / 0	holds with imported ingredients	the pair rule P = η(u_s, u_r)/η(u_s, u_−s) (translation by the AI), the fourth side, fair first outcome, mirror
T4 with q = ½ (probability (η/η_opp)^½)	0.1659	0.4142	2.164784	—	fails	—
T4 with q = 2	0	0.5	2.828427	—	fails on C2	—
T5 common record read locally (Bell 1964)	0.2071 on the grid, 0.2105 continuous	0	2 / 2	0 / 0	fails, as registered (control)	—

What the checks show.

- T2: with P(A = u) = ½ and the mirror, E = 2f(−c) − 1 whenever f(c) + f(−c) = 1 (sympy). The mirror gives the perfect anticorrelation at θ = 0 and the form of the correlation, not the curve: with a law linear in the angle it gives Bell's straight line and S = 2. **In this model, once the mirror is given, going beyond 2 depends on the law with which one end is read**, in line with section 4.1 (the bound is set by the local reading).
- T3: with f = cos²g the Fisher information is 4g′², so equal distinguishability for equal rotations means g linear; f(0) = 1 and f(π) = 0 make κ a half-integer, monotonicity selects κ = ½, that is cos²(θ/2), the half angle, without SU(2) put in. Counting the fractions without their noise selects instead the law linear in the angle (Bell's line). This is not new: Wootters (1980–81, not read directly here; known through Wootters 2013), Summhammer (quant-ph/9910039, eq. 6), Brukner and Zeilinger (Found. Phys. 39, 677, 2009, with a quadratic measure of information assumed). It needs copies independent of each other, whereas the author's premise of 13 October speaks of correlated copies.
- T4: the forms Q = Σ_{i<j} x_i x_j have eigenvalues (1, −½, −½) for three sides and (3/2, −½, −½, −½) for four, signatures (1, 2) and (1, 3). For null directions the normalised product gives sin²(θ/2) (deviation $4\cdot10^{-16}$), and writing a light direction as $2\psi\psi^{T}$ rotates ψ by θ/2. With three sides the readings lie in a plane; with four sides they cover the three directions and T4b gives −a·b in space. The computation uses real directions only: complex numbers are not needed for these correlations and the fourth side does not impose them; they enter only if one asks to write a light direction as the product of a two-component object ($S^2 \cong CP^1$, the spinor–null-direction correspondence of Penrose and Rindler, quoted from memory, not checked against the source here). The pair rule is the Born rule of a qubit written on light directions: a translation by the AI, not new.
- T1: with χ ≠ π the correlation depends on the sum of the angles (C4 fails); k = 1 gives −cos 2θ in the x–z plane (the photon curve) and breaks isotropy in space; p ≠ 2 fails C1, with S = 3.469182 (p = 3) and 3.771236 = 4·(2√2/3) (p = 4): these weights are not quantum mechanics, so this does not violate the quantum Tsirelson bound.
- One registered expectation did not come true: for T4 with q = 2 (probability (η/η_opp)²) I expected a failure of C1; instead E = f(−c) − f(c) = −c exactly and the marginal of B is (1 − c²)/2, up to 0.5 at θ = π/2 (C2 fails).
- No translation that holds deviates from −cos θ: this is a compatibility with the experiments (values read in the articles listed in the Level-0 inventory, Sakai from the abstract only; Hensen et al. 2015, S = 2.42 ± 0.20; Storz et al. 2023, 2.0747 ± 0.0033; Poh et al. 2015, 2.82759 ± 0.00051; Sakai et al. 2006, proton pairs in the singlet, 2.83 ± 0.24 ± 0.07), not a prediction.

Reading for MM. The sentence of the common introduction on the mirror-like reciprocity of the two ends holds as a reason for the anticorrelation and the form of the correlation, not for the curve: the curve comes from the law of one end.

Imported in every translation that holds: the fair outcome of the first end read (v), the mirror as an update rule (iv), and the three directions as directions of the end (isotropy or the fourth side) (iii).

Verification. Astra (Codex, gpt-6-luna): 11 remarks (counts in the notes on the reports, in the registrations), one of grade A (the protocol wrote the T4 rule as η/2, true only in normalised coordinates; the code uses the ratio, which is now the declared ingredient); Argon (Gemini): 9 remarks of grade B/C. Both: «ready with corrections»; no remark changed an outcome. The corrections are in the outcome note v3 (`documenti/working_notes_it/07_singoletto_NOTA_ESITO_LIVELLO1_v3.md`).

5. Placement in the Local Friendliness taxonomy (04_verbale_loop)

Cavalcanti and Wiseman (2021) define **Local Agency** as "the only relevant events correlated with an intervention are in its future light cone"; it replaces Locality and No-Superdeterminism together. They split the relativistic causal arrow into a **temporal causal arrow** and **relativistic causality**.

	placement of August 2026 (status report 2.7–2.27)	**placement from 13 October 2026**
Absoluteness of Observed Events	kept in the verbale of 24 August; revised on 25 August ("the object is absolute, the reading partial")	**abandoned in the original definition.** A reading made inside a laboratory that leaks no copies (the friend in an extended Wigner scenario) is not absolute; readings with enough copies agree (25 August revision)
No-Superdeterminism	kept	**violated** by future-input dependence, as in the model of 4.4 (not a conspiracy of initial conditions)
Local Agency	violated	violated (follows from the line above)
temporal causal arrow	kept	**abandoned** (the loop)
relativistic causality	abandoned ("realist non-local")	**retained as a programme assumption** (the author's choice, "variant A", 13 October 2026). The pair model implements local path mediation; whether the cited principle is satisfied needs a separate causal argument. If causation were taken as transitive, the chain choice → reading → source → other reading would link space-like separated events

This is a placement, not a result. Two-time formulations that reproduce all standard probabilities are empirically equivalent to quantum mechanics in that domain; for MM this equivalence is not yet shown.

6. Corrections to texts already published

Versions 2.7–2.18 of the status report contain the August placement texts (versions 2.17, 2.19 and 2.20 were not published): in Italian from 2.7, and in English as `QUANTUM_PLACEMENT.md` from 2.9. The archive `MM_v8_addendum_26082026.zip` holds `PROPOSTA_scrivere_e_fondere.md`. Versions 2.21–2.27 carry these texts inside the inherited archive `MM_base_pubblicata_2.18.zip`. The following statements are corrected:

1. **"The sense is the one Bell's theorem is about"** (QUANTUM_PLACEMENT §0). "Non-local" now means pregeometric (section 3, terminology). Bell correlations are carried by the loop with local path mediation (section 5).
2. **The conditional theorem** ("no fact determines which outcome is written") is withdrawn in the stated form. For bits r and m with an initially blank record m = 0, the reversible map U(r, q, 0) = (r, q, r) writes an antecedent relational fact into a record without merging anything. The selection of the outcome is now a separate working assumption (W2), to be formalised.
3. **"The conservation axiom coincides with the purification axiom of Chiribella, D'Ariano and Perinotti"** (and the book "inherits their theorems"). It is a resemblance, to be shown formally; their reconstruction uses further axioms (PRA 84, 012311, 2011).
4. **"Three debts shown by others to be unavailable."** They are conditional constraints, not proofs of impossibility:
5. weight → frequency: Schlosshauer and Fine, Found. Phys. 35, 197 (2005), analyse the assumptions of derivations;
6. non-locality: Bell and Local Friendliness constrain explanations that keep given assumptions;
7. a stable probability measure: the same family as weight → frequency.
8. **The August placement table** is replaced by section 5.

The book NUMERI (v119, §[01435]) says the Born exponent follows "from these premises alone". That claim is withdrawn: the argument also assumed linear dynamics. Strong positivity supplies a Hilbert-space representation, but this deposit does not derive the required dynamics or Born frequencies from it.

7. What is not claimed, and what is open

Not claimed:

- a derivation of the Born rule without assumptions: in sections 4.1–4.5 quadratic weights, local Born responses and suitable measures over futures are assumed; section 4.6 (2.48.1) obtains the Born rule from counting copies of equal modulus only with the declared assumptions H-es, H-eq and H-add (and H-cont for irrational ratios), with a construction analogous to Zurek's in which H-eq is a declared premise, and the physical necessity of H-eq is not derived;
- a derivation of Tsirelson's bound beyond the published theorems;
- a derivation of the singlet curve $-\cos\theta$ without imported ingredients: in section 4.7 (2.48.2) it holds in three translations, each with the ingredients listed there (law of one end, mirror update, fair first outcome, three directions);
- an experimental distinction from standard quantum mechanics;
- mathematical priority.

Open:

1. a rule that selects one outcome, with a measure over futures that gives Born. Section 4.5 shows that non-Born weights make consistent histories arbitrarily improbable in the pair model; turning this into a reason for the Born weight needs a postulate (stated there) and a general argument;
2. why the returning copy is the complex conjugate of the outgoing one (in the model it follows from defining the return with the adjoint step, section 4.6; a physical reason remains open), and why copies of equal modulus count the same (H-eq, which in section 4.6 is the author's premise U1);
3. a locally mediated model beyond a singlet pair: several particles, networks, adaptive choices;
4. why the hidden parameter is not accessible, and a causal argument for relativistic causality;
5. the value of $B(p)$ for $1 < p < 2$, and its dependence on dimension.

8. Reproduction

- `python codice/run_all.py --short` runs everything except the three long programs (about 15 minutes).
- `python codice/run_all.py` runs everything (about 2 hours; `tsirelson.py` alone takes about 80 minutes).
- `python codice/run_all.py --compare-only` re-compares an existing `uscite/`.

Block 06_born_conteggio (2.48.1) runs in both modes and takes a few seconds. Its reference log `risultati/06_born_conteggio/log_K1_K5.txt` is the registered log without the three timing lines (real, user, sys) added by the shell's `time` command, which are not program output; the registered log is kept unchanged in `documenti/registrations_original/06_born_conteggio/02_livello1/`.

Block 07_singoletto (2.48.2) runs in both modes and takes under a minute. It runs `calcoli_livello1_c1.py`, the corrected program registered before its run (correction 1, section 4.7); the first registered version is kept in `documenti/registrations_original/07_singoletto/02_livello1/`.

Each block runs in `uscite/<block>/` and is compared with `risultati/<block>/` under the rules in the docstring of `run_all.py`:

- standard output only, with the expected numpy division warnings removed from the references;
- running times masked only where they are running times (end-of-line parentheses, "FINE in", "calcolate in"), plus start dates and the numpy version;
- floats equal within a relative tolerance of 10^{-9} plus an absolute 10^{-12}; integers (and integer-written numbers in logs), booleans, infinities and NaN exactly equal;
- JSON compared without its running-time fields ("secondi", "sec"), right after each program and again at the end; figures not compared;
- each program's exit code stored; a missing output, a missing status or a non-zero exit is a failure;
- `controllo_d3.py` checked as a matching prefix with exactly 8 starts and a final line.

"Matching" means agreement with the stored references under these rules; it does not certify the references themselves.

Requirements: Python 3, numpy, scipy, matplotlib (tested with Python 3.13, numpy 2.4.6, scipy 1.18.0).

The reference for `tsirelson.py --rapido` was regenerated with the final program. The working quick run had been made before the declared changes (faster marginal test, added seeds) and is kept as `risultati/01_tsirelson/log_rapido_codice_precedente.txt`. The registered outcomes come from the full run.

The programs are the working files, unchanged, with one exception: `controllo_d3.py` reads the number of starts from the environment variable PARTENZE, because the working run was stopped by hand after 8 starts.

9. Provenance

block	registration (UTC, 13 October 2026)	SHA-256 of the registered file
01 Tsirelson	15:41:53	6e0277aa…
02 two slits, first run	17:48:40	82a5526d…
02 two slits, second run	17:51:58	07922f3e…
03 loop consistency	18:08:51	683fcfba…
04 placement, author's choice of variant A	19:14:47	d252047d…
05 pair model	19:17:49	d7711e20…
R2 second round	19:30:20	b5c1550f…
R3 self-consistency	20:09:57	a9ecca20…

Block 06, new in 2.48.1 (UTC, 25–26 October 2026; registers in `documenti/registrations_original/06_born_conteggio/`):

step	registration (UTC)	SHA-256 of the registered file
author's premises and candidate sources	25 Oct 23:49:56	f1379750…, 7e8dcb0b…
fingerprints of the 22 sources	25 Oct 23:53:43	ac22ebb3…
Level-0 inventory and the four readings	26 Oct 00:10:09	9664c7e7… (inventory)
Level-1 protocol	00:11:30	2acc00b8…
first verification and corrections 1	00:31:55	be792a27… (protocol correction)
second verification, correction 2 and program	00:37:02	46c25746… (correction 2), 763b4d80… (program)
results of K1–K5	00:37:28	8fc342c1…
K5bis, added after the run, registered before running it	00:37:50	0150d11b…
outcome note	00:38:37	73ef4a06…

Block 07, new in 2.48.2 (UTC, 29 October 2026; registers in `documenti/registrations_original/07_singoletto/`):

step	registration (UTC)	SHA-256 of the registered file

author's premises and Level-0 inventory v0	13:01:10	69f84bb2…, f7d986c8…
fingerprints of the sources (7 on disk, 22 downloaded)	13:01:10, 13:04:32	ee53132d…, ed075ebf…
inventory v1, Level-1 protocol and program	13:24:30, 13:24:31	cb7ef867…, 76a0bcbb…, 164382b9…
correction 1 and corrected program	13:25:09	57dcd022…, 43fcd771…
outcome note v1	13:26:55	122795f3…
Astra G1 report and note; outcome note v2	13:32:22–13:33:20	30e4d6b9…, 94cfcd64…, 56c3e141…
Argon G1 note; outcome note v3	13:53:02	96489c61…, a660457a…

For block 07 the 29 source PDFs, their extracted text and the full messages to the verifiers are left out as well; the message to Argon is the same material that was sent to Astra.

Left out of the package: the 22 source PDFs and their extracted text (not redistributed; only their fingerprints), and the messages to the verifier, which contain extracts of the sources.

The full fingerprints and times are in `documenti/registrations_original/*/REGISTRO.txt`, next to the original prediction files, unchanged. A fingerprint with a local time documents the content; it does not by itself certify the time. The Italian working notes after verification are in `documenti/working_notes_it/`.

10. Files

- `codice/<block>/`: the programs; `codice/run_all.py`
- `risultati/<block>/`: reference logs, JSON files and figures
- `documenti/REGISTERED_TESTS.md`: every registered criterion with its outcome
- `documenti/registrations_original/`: predictions and registers as written
- `documenti/working_notes_it/`: Italian notes after each verification round
- `LEGGIMI_IT.md`: Italian summary
- `codice/06_born_conteggio/`, `risultati/06_born_conteggio/`: block 06, new in 2.48.1
- `documenti/registrations_original/06_born_conteggio/`: the author's premises, fingerprints of the sources, the readings, inventory, protocol, corrections, verification reports and registers of block 06
- `documenti/working_notes_it/06_born_conteggio_NOTA_ESITO_LIVELLO1_v1.md`: Italian outcome note of block 06
- `IMPRONTE.sha256`: fingerprints of every file

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